

No. of Functional Features Included in the Solution

Date	24 June 2026
Team ID	SWTD-2026-3249
Project Name	OptiCrop - Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the OptiCrop solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S. No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	ML-Based Crop Recommendation Engine	Predicts the optimal crop and expected yield using soil, weather, and historical data.	Core Logic Service (ML)	Done	1.0
2	Real-Time Irrigation & Fertilizer Advisory	Generates weather-aware, field-specific irrigation and fertilizer dosage recommendations.	Core Logic Service (ML)	Done	0.75
3	Farmer Mobile App (Offline-First)	Flutter-based mobile app allowing farmers to view recommendations with limited connectivity.	Presentation Layer	Done	0.75
4	Agronomist Web Dashboard	React.js dashboard for agronomists/admins to monitor farms, data, and recommendations.	Presentation Layer	Done	0.5
5	SMS Alerts for Non-Smartphone Users	Sends key advisories via SMS using Twilio for farmers without smartphone access.	External API Integration	In Progress	0.5
6	Geospatial Field Mapping	Lets farmers mark field boundaries and view location-based insights via Google Maps API.	External API Integration	Done	0.5
7	User Authentication & Role Management	Secure login and role-based access for farmers, agronomists, and admins.	Auth Service	Done	0.5

8	Continuous Learning from Farmer Feedback	Refines recommendation models each season using farmer-reported outcomes.	Core Logic Service (ML)	Pending	0.5
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Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	6
Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	6

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Farmer Mobile App, Agronomist Web Dashboard
2	Backend / Logic	2	ML Crop Recommendation Engine, Irrigation & Fertilizer Advisory
3	Database / Storage	1	Farm, Crop & Sensor Data Storage
4	API / Integration	2	SMS Alerts (Twilio), Geospatial Field Mapping (Google Maps)
5	Security / Authentication	1	User Authentication & Role Management